

Output and Costs

Producer Theory

- Firm's decisions:
 - What goods and services to produce
 - How to produce (what factors of production to use)
 - How to price the products
- Goal of the firm is to maximise profits

$$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$$

Producer Theory

- *Total revenue* is the amount that a firm receives from the sale of goods and services and is calculated as the quantity sold multiplied by the price paid for each unit:

$$\text{Total revenue} = \text{Quantity} \times \text{Price}$$

- *Total cost* is the amount that a firm pays for inputs used to produce goods or services.

Fixed and Variable costs

- A firm's total cost is defined as:
$$\text{Total costs} = \text{Fixed costs} + \text{Variable costs}$$
- *Fixed costs* are costs that do not depend on the quantity of output produced.
 - One-time, upfront payments before production begins, like buying equipment.
 - Ongoing payments, like monthly rents.
- Fixed costs are constant as quantity increases.
- Even if a firm produces nothing, it still incurs a fixed cost.

Fixed and Variable costs

- *Variable costs* are those that depend on the quantity of output produced.
 - Includes raw materials as well as labor costs.
- Total variable costs (TVC) increase with each additional unit produced.
- If a firm produces nothing, variable costs are zero.

Time Frames for Decision Making

Short-run: a time-frame in which the quantity of at least one factor of production is *fixed*

Long-run: a time-frame in which the quantity of all factors of production can be *varied*

Short-Run

Short-run: a time-frame in which the quantity of at least one factor of production is fixed

For most firms

- *capital, land and entrepreneurship are fixed factors of production (usually called a firm's plant) e.g. factories, machinery*
- *labor is the variable factor of production*
- In the short run, to increase output a firm must increase the quantity of the variable factor of production
- These decisions are *easily reversed*

Long-Run

Long-run: a time-frame in which the quantity of all factors of production can be varied

- Time-frame in which the quantity of all factors of production can be varied
- In the long-run, the firm can change its plant and/or hire more labor to increase output
- These decisions are *not easily reversed*

Short-Run vs Long-Run

- The only costs that influence a firm's decisions are
 - The short-run cost of changing its labor inputs
 - The long-run cost of changing its fixed resources/firm's plant

Short-Run Technology Constraints

Three measures of output:

1. Total Product: The total output that can be produced by a given quantity of labor
2. Marginal Product: The increase in total product resulting from a one unit increase in labor, with all other inputs remaining the same
3. Average Product: The total product divided by the quantity of labor employed (measures worker efficiency)

Output & Costs

Short Run Technology Constraints

Notice the patterns:

1. As no. of labor employed increases, total product increases
2. As no. of labor employed increases, marginal product at first increases then begins to decrease
3. As no. of labor employed increases, average product at first increases then decreases

Table 10.1

Total Product, Marginal Product and Average Product

	Labour (workers per day)	Total product (jumpers per day)	Marginal product (jumpers per worker)	Average product (jumpers per worker)
A	0	0		
			4	
B	1	4		4.00
			6	
C	2	10		5.00
			3	
D	3	13		4.33
			2	
E	4	15		3.75
			1	
F	5	16		3.20

Product Curve- Marginal Product Curve

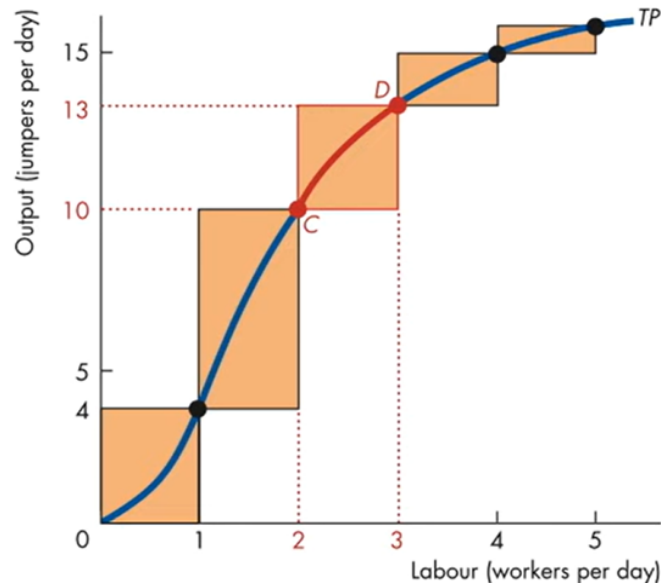
Marginal Product (MP) can be measured by the slope of the Total Product Curve (TP) and it shows the output from an additional labour

The shapes of the product curve represents:

Increasing Marginal Returns initially

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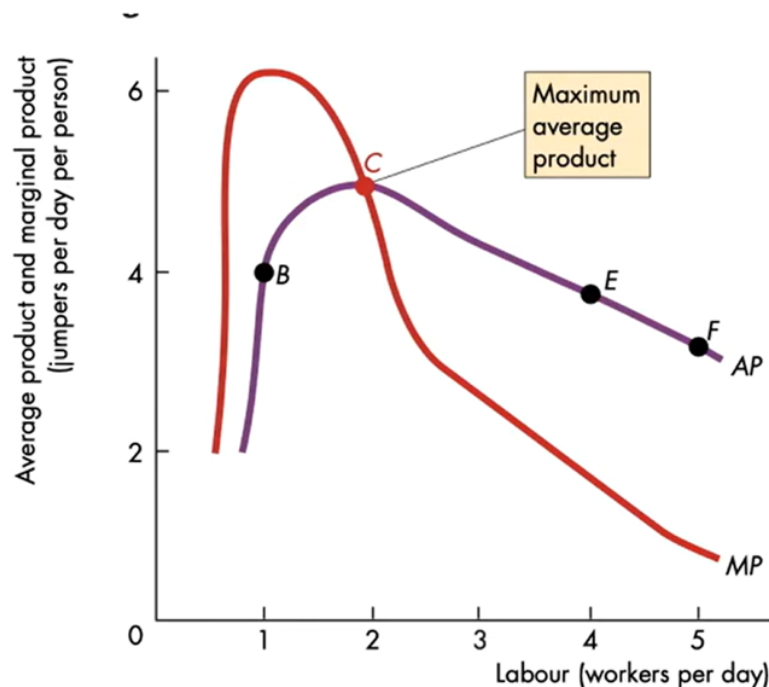
Diminishing Marginal Returns later



Law of diminishing returns- So this proves, as firm uses more of a variable factor of production, with a given quantity of fixed factor of production, the marginal product of the variable factor eventually diminishes

Product Curve- Average Product Curve

- Average Product (AP) hits its maximum when it cuts the Marginal Product (MP) curve
- AP is increasing when for the number of workers MP curve exceeds the AP curve
- AP is decreasing when for the number of workers MP curve falls below AP curve



Short-run costs

In the short-run, in order to increase output, a firm must increase the quantity of labor – which means it must increase its cost

Three measures of cost:

1. *Total Cost*: Total cost of all resources/factors of production used in the production process
2. *Marginal Cost*: Increase in total cost that results from a one-unit increase in total product or output
3. *Average Cost*: Total cost per unit of output produced

Short-run costs

- *Total cost* is the cost of all resources/factors of production used in a production process

It can be divided into two parts:

- *Total Fixed Cost*: Cost of the firm's fixed inputs/factors. In the short run, total fixed cost does not change as output changes (e.g. factory rent, cost of renting machinery)
- *Total Variable Cost*: Cost of the firm's variable inputs/factors. Total variable cost changes as output changes (e.g. cost of labor)

$$TC = TFC + TVC$$

Costs of production

- When a firm increases its output by adjusting its use of inputs, it incurs the costs associated with that decision.
- The relationships between output and costs are:

$$\text{Average fixed cost (AFC)} = \frac{\text{Fixed cost}}{\text{Quantity}}$$

$$\text{Average variable cost (AVC)} = \frac{\text{Variable cost}}{\text{Quantity}}$$

$$\text{Average total cost (ATC)} = \frac{\text{Total costs}}{\text{Quantity}} = \text{AFC} + \text{AVC}$$

$$\text{Marginal cost (MC)} = \frac{\Delta \text{ total cost}}{\Delta \text{ quantity}}$$

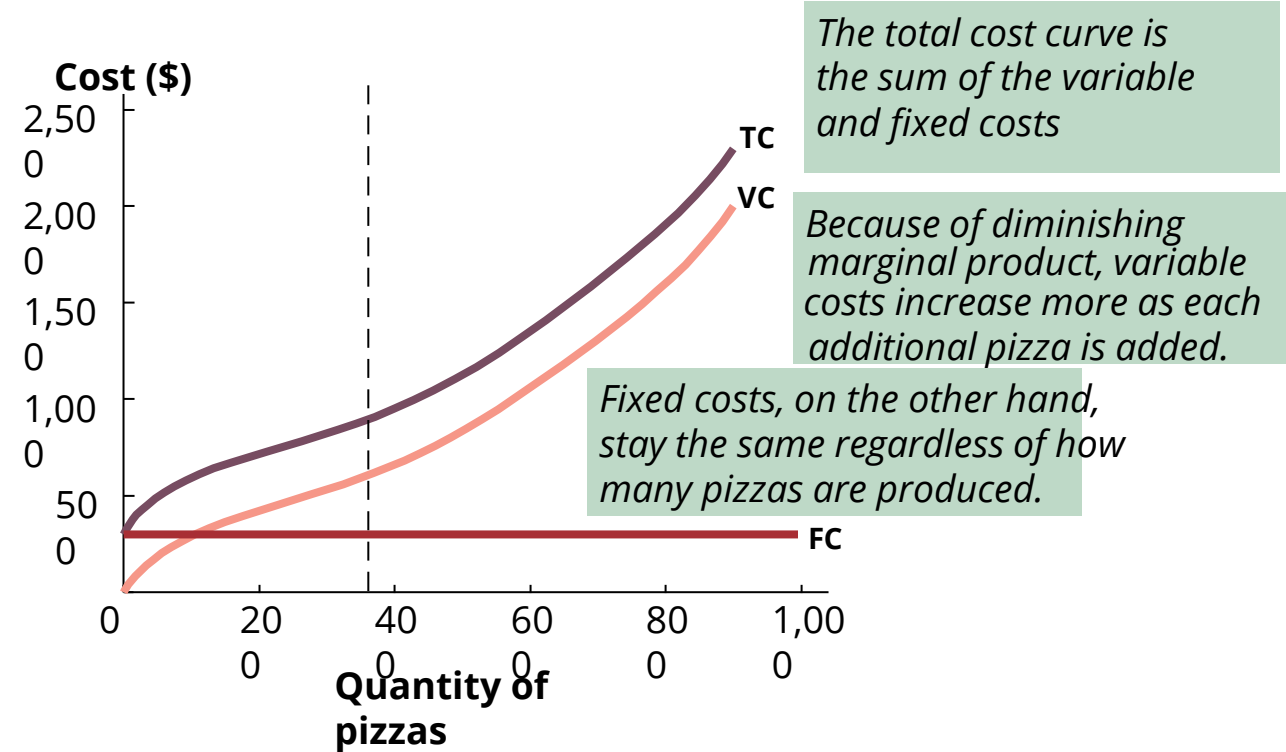
Costs of production

The table below provides the production and costs of a pizza joint that requires a lease for \$300 and wages for workers at \$200 each.

** All values may not be correct in this table. Please follow class lecture.*

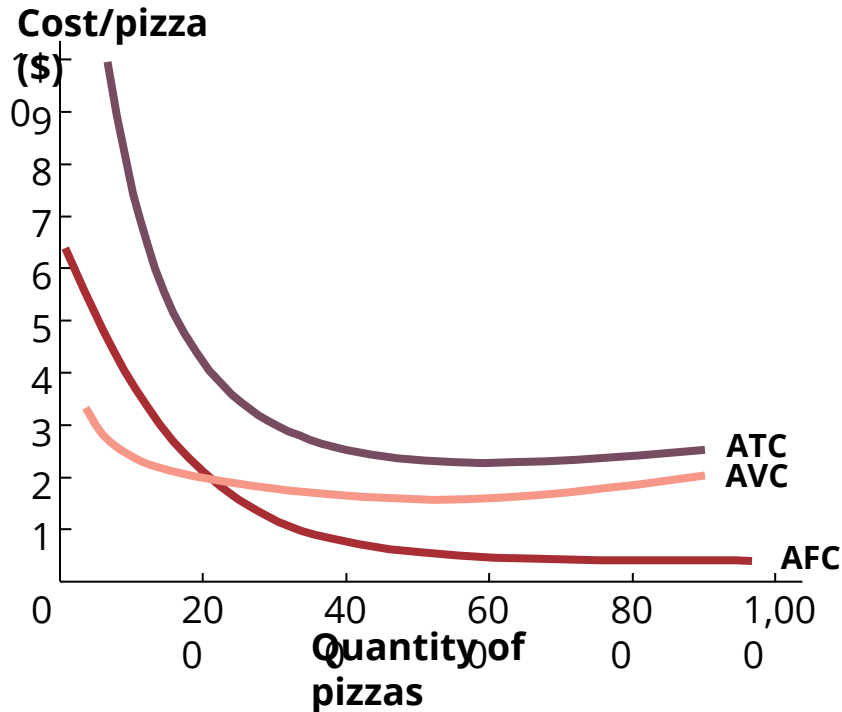
Labor (# workers)	Total production (# pizzas)	Fixed costs (\$)	Average fixed costs (\$/pizza)	Variable costs (\$)	Average variable costs (\$/pizza)	Total costs (\$)	Average total costs (\$/pizza)	Marginal product (# pizzas)	Marginal cost (\$/pizza)
0	0	300		0		300		—	—
1	50	300	6.00	200	4.00	500	10.00	50	4.00
2	180	300	1.67	400	2.22	700	3.89	130	1.54
3	360	300	0.83	600	1.67	900	2.50	180	1.11
4	480	300	0.63	800	1.67	1,100	2.30	120	1.61
5	575	300	0.52	1,100	1.74	1,300	2.26	95	2.11
6	660	300	0.45	1,200	1.82	1,500	2.27	85	2.35
7	735	300	0.41	1,400	1.90	1,700	2.31	75	2.67
8	800	300	0.38	1,600	2.00	1,900	2.38	65	3.08
9	855	300	0.35	1,800	2.11	2,100	2.46	55	3.67
10	900	300	0.33	2,000	2.22	2,300	2.55	45	4.44

Cost curves



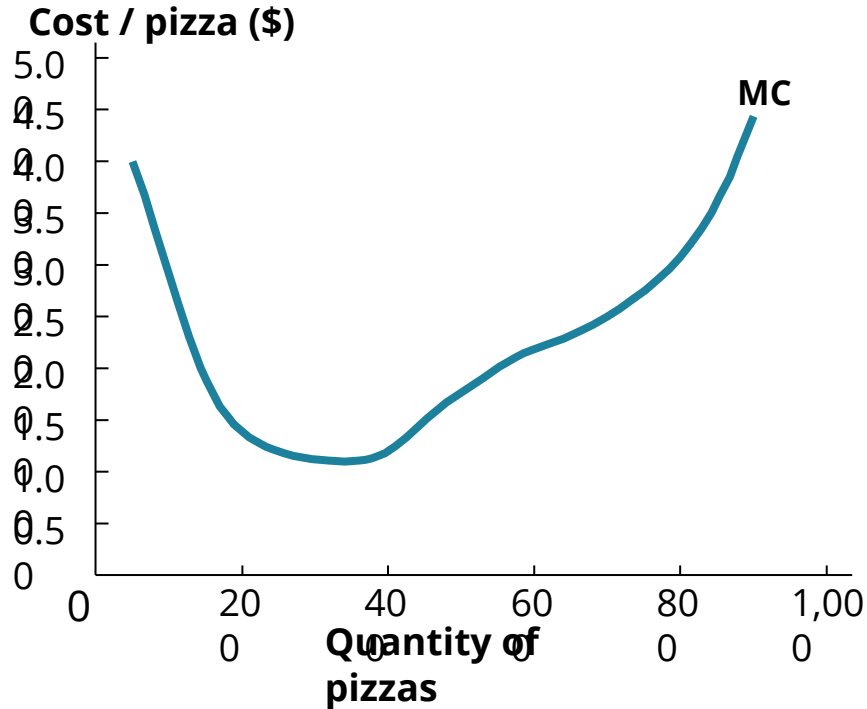
- The VC curve initially becomes less steep, reflecting the increasing marginal product of the first few employees.
- As the principle of diminishing marginal product kicks in, the variable cost curve gets gradually steeper.
- Adding FC and VC yields TC.

Cost curves

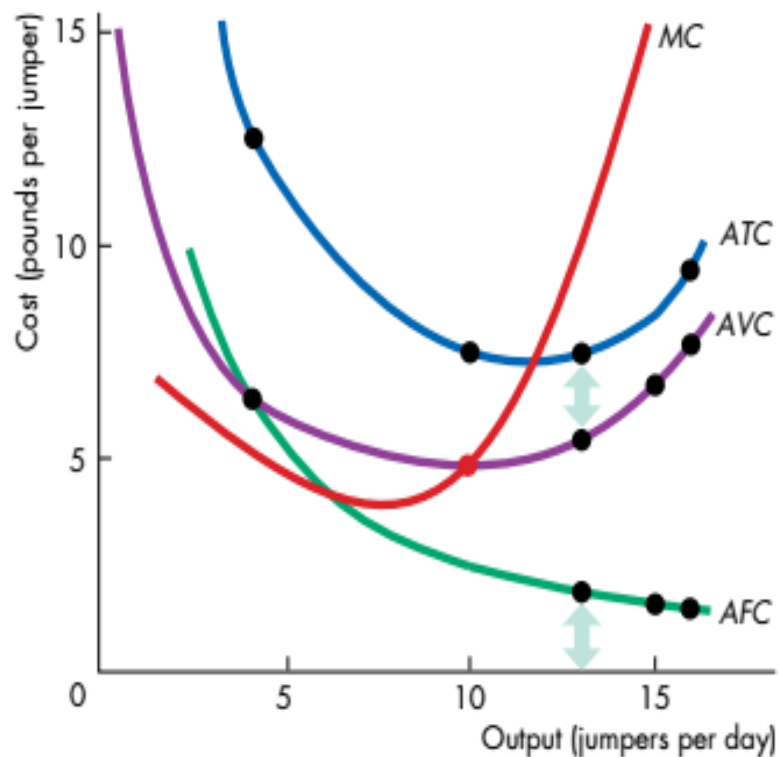


- AFC trends downward.
 - Same cost spread out over more units of output.
- AVC is U-shaped.
 - First decreases and then increases, reflecting the marginal product of inputs.
- ATC curve is U-shaped.
 - Decreases in AFC and increases with AVC.

Cost curves



- The MC curve is U-shaped and is the inverse shape of the marginal product curve.
 - Marginal cost decreases at low outputs because of economies from greater specialization. It eventually increases because of the law of diminishing returns.
 - Each additional unit of labor produces a successively smaller addition to the total output
 - This means to get the same addition to output, even more workers are required. Therefore, marginal cost (cost of an additional unit of output) also increases eventually
 - As marginal product of labor initially increases, MC decreases and as marginal product diminishes, the MC increases.



Marginal cost is calculated as the change in total cost divided by the change in output. When output increases from 4 to 10, an increase of 6, total cost increases by £25 and marginal cost is $\text{£}25 \div 6$, which equals $\text{£}4.17$. Each average cost concept is calculated by dividing the related total cost by output. When 10 jumpers are produced, *AFC* is $\text{£}2.50$ ($\text{£}25 \div 10$), *AVC* is $\text{£}5$ ($\text{£}50 \div 10$), and *ATC* is $\text{£}7.50$ ($\text{£}75 \div 10$).

The graph shows the marginal cost curve and the average cost curves. The marginal cost curve (*MC*) is U-shaped and intersects the average variable cost curve and the average total cost curve at their minimum points. Average fixed cost (*AFC*) decreases as output increases. The average total cost curve (*ATC*) and average variable cost curve (*AVC*) are U-shaped. The vertical distance between these two curves is equal to average fixed cost, as illustrated by the two arrows.

Shifts in the cost curves

A firm's cost curves shift in the short-run due to two main factors:

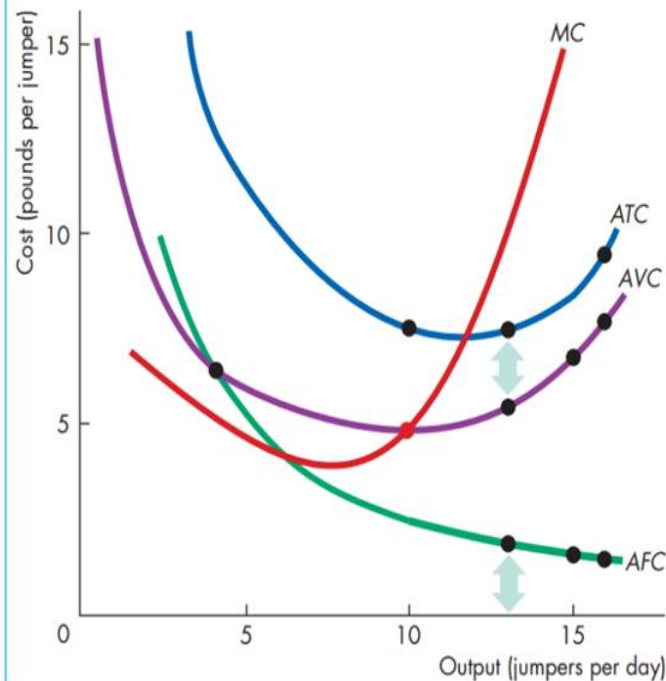
1. **Technology:** Any technological advancement that makes the production process more efficient will shift product curves upwards and cost curves downwards (using the same resources, we can produce more products at a lower cost)
2. **Prices of factors of production:** An increase in the price of a factor of production increases costs and shifts the cost curves upwards.

Figure out on your own:

Which curves will shift if rent increases?

Which curves will shift if wage rate increases?

Figure 10.5 Marginal Cost and Average Costs



Summarizing costs

The following summarizes various costs.

Cost	Description	Calculation
Total cost (TC)	The amount that a firm pays for all of the inputs (fixed and variable) that go into producing goods and services	$TC = FC + VC$
Fixed cost (FC)	Costs that don't depend on the quantity of output produced	—
Variable cost (VC)	Costs that depend on the quantity of output produced	—
Average fixed costs (AFC)	Fixed costs divided by the quantity of output	$AFC = FC / Q$
Average variable costs (AVC)	Variable costs divided by the quantity of output	$AVC = VC / Q$
Average total costs (ATC)	Total costs divided by the quantity of output	$ATC = TC / Q$
Marginal cost (MC)	The additional cost incurred by a firm when it produces one additional unit of output	$MC = \Delta TC / \Delta Q$

Long-run costs

- In the long-run, all factors of production can be varied
- In the long-run, all costs are variable costs
- In the long-run, if a firm wants to increase its output, it can
 - Hire more labor
 - Increase quantity of raw materials or other inputs
 - Build a new plant/factory/acquire or rent new space
 - Buy or rent new machineries

Long-run costs

- Long-run costs depend on the firm's production function
- A production function describes the relationship between the maximum output attainable and quantities of different inputs used in the production process (usually labor and capital)
- Some production processes use more labor and less capital while others use lower quantities of labor and more capital –

Long Run Costs

- The table shows total product for different combinations of labor and capital
- Quantity of capital is defined by plant size
- For a given quantity of capital, as we increase labor, the marginal product of labor diminishes
- For a given quantity of labor, as we increase capital, the marginal product of capital diminishes

The Production Function

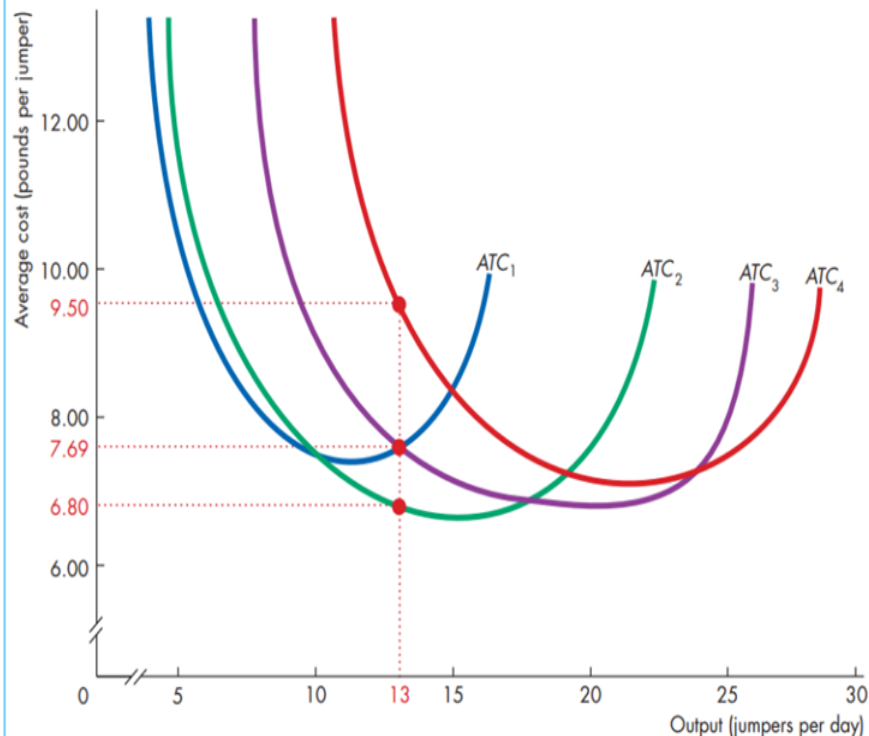
Labour (workers per day)	Output (jumpers per day)			
	Plant 1	Plant 2	Plant 3	Plant 4
1	4	10	13	15
2	10	15	18	20
3	13	18	22	24
4	15	20	24	26
5	16	21	25	27
Knitting machines (number)	1	2	3	4

The table shows the total product data for four quantities of capital – four plant sizes. The bigger the plant size, the larger is the total product for any given amount of labour employed. But for a given plant size, the marginal product of labour diminishes as more labour is employed. For a given quantity of labour, the marginal product of capital diminishes as the quantity of capital used increases.

Long Run Costs

- In the long run, a firm chooses its plant size depending on the output that it plans to produce.
- This is because the average total cost of producing a given output depends on the plant size.
- Suppose the firm wants to produce 13 jumpers in a day
- With one plant (ATC₁), ATC = £ 7.69
- With two plants (ATC₂), ATC = £ 6.80
- With three plants (ATC₃), ATC = £ 7.69
- With four plants (ATC₄), ATC = £ 9.50
- The economically efficient plant size for producing a given output is the one that has the lowest average total cost (in this case ATC₂).

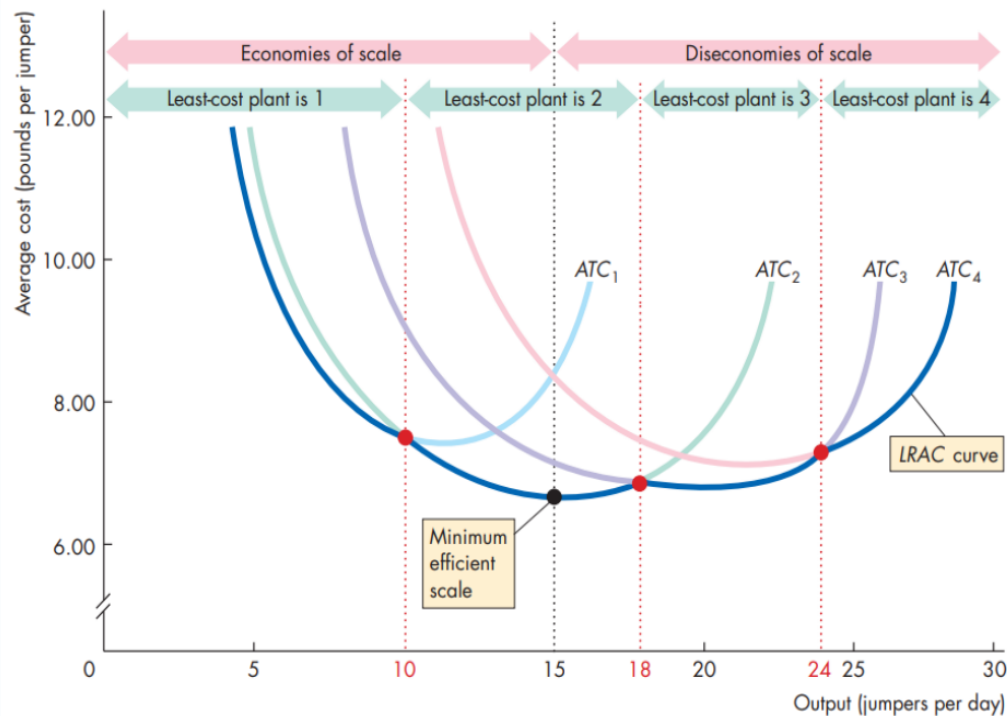
Figure 10.8 Short-Run Costs of Four Different Plants



Long Run Costs

- When a firm is producing a given output at the least possible cost, it is operating on its long-run average cost curve.
- The long-run average cost curve is the relationship between the lowest attainable average total cost and output when both the plant size and labor are varied
- Long-run average cost curve is made up from the lowest ATC for each output level – the least costly way of producing a given level of output
- The deep blue curve is the LRAC curve

Figure 10.9 The Long-Run Average Cost Curve



Economies & Diseconomies of Scale

Economies of Scale: As we produce more, our average cost fall – this happens because of greater specialization of both labor and capital

- The LRAC curve shows that this production process experiences economies of scale for outputs up to 15 jumpers a day.

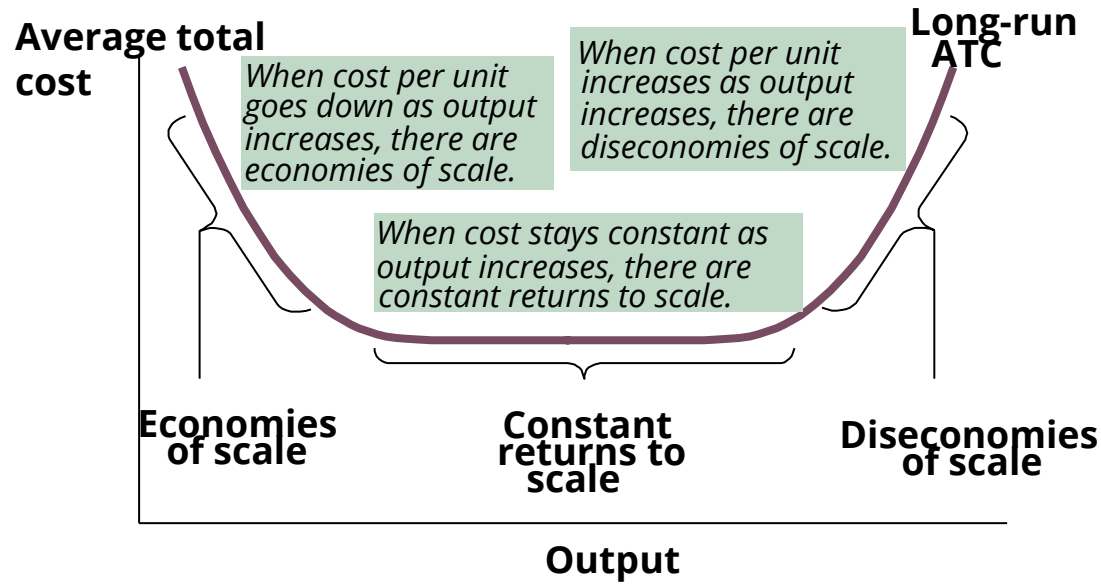
Diseconomies of scale: As we produce more, our average cost rises (due to technology used by firm and other firm-related aspects)

- When diseconomies of scale are present, the LRAC curve slopes upward.

Constant returns to scale are features of a firm's technology that keep average total costs constant as output increases.

- When constant returns to scale are present, the LRAC curve is horizontal.

Economies and diseconomies of scale



Practice Problem: Costs

Fill in the table, assuming that a lease for a building is \$300 and workers' wages are \$100 each.

Labor (# workers)	Total production (# pizzas)	Fixed costs (\$)	Average fixed costs (\$/pizza)	Variable costs (\$)	Average variable costs (\$/pizza)	Total costs (\$)	Average total costs (\$/pizza)	Marginal cost (\$/pizza)
0	0							–
2	45							
4	100							
6	135							
8	150							
10	165							

Summary

- *Key concepts:*
 - Fixed and variable costs
 - Short-run vs Long-run
 - Short-run technology constraints
 - Short-run costs
 - Long-run costs
 - Economies and Diseconomies of Scale
- *Reading: Chapter 10 of “Economics” by Parkin, Powell, Mathews 6th Ed.*